MUHAMMAD ZAHID	BT401 Genetic Resources & Conservation
Current Papers	Fall 2020

Questions:

1. **Role of botanical garden in conservation (PPT 80-81)**
   Importance of Botanical garden
   1- Enjoyment
   2- Economic
   3- Scientific research
   4- Conservation
   5- Climate Change
2. **Who is responsible for botanical garden?**
   Botanical gardens are often run by universities or other scientific research organizations, and often have associated herbaria and research programmers in plant taxonomy or some other aspect of botanical science
3. **Historical background of Ramsar convention**
   History: Ramsar is one of the global inter-governmental environmental agreements. The treaty was negotiated in1960s by countries and NGOs. To avoid the increasing loss and degradation of wetland habitat for migratory water birds. In 18 nations meeting it was adopted in the Iranian city of Ramsar On 2nd February 1971. Came into force in 21st December 1975
4. **Disadvantage of DNA bank**
   Advantages
   DNA banking is an efficient, simple and long-term method to conserve the genetic information.
   Disadvantages
   There are problems with subsequent gene isolation, cloning and transfer of DNA back to a plant and it currently does not allow the regeneration of the same genotype as the original sample.
5. **In vivo gene bank**
   Generally, plant seeds, vegetative propagules are used for storage for long time. The whole plants are preserved. This type of conservation strategy is called in vivo gene bank. In this approach, conservation method of storage is used for preservation of plant genetic resources
6. **Central goal of any zoo**
   Unfortunately, not all zoos are scientific in nature, and extreme controversy has arisen regarding how the animals are treated. Suffice to say, regulation is necessary to ensure proper care. Conservation (not exploitation) should always be the central goal behind any legitimate zoo. Zoos provide the opportunity for people to see a glimpse of this side of nature. Zoo plays important role in conservation of many threatened/endangered species.
7. **The full form of EEZ**
   Exclusive economic zone
8. **Application of permit to import of plant genetic resources.**
   (a) the kind and quantity of plant or plant material;
   (b) the country and locality of origin;
   (c) destination;
   (d) the name and address of the consignor and the consignee;
   (e) means of transport;
   (f) the prescribed port or point of entry;
   (g) the prescribed for which the plant or plant material is proposed to be imported e.g., consumption propagation or processing.

Muhammad Zahid (Bhaweli Lakha)

9. **Extreme weather effect of invertebrate genetic resource**
Extreme weather events such as heat waves, droughts and floods - which are predicted to increase in frequency due to climate change – are often followed by pest outbreaks. Among other contributing factors, these outbreaks can occur because the extreme event eliminates or weakens a pest's natural enemies.

10. **What is the role of cryoprotectant in cryopreservation?** **3**
A **cryoprotectant** is a substance used to protect biological tissue from freezing damage. Due to ice crystal formation cell membrane and cell integrity can ruptured. **cryoprotectants** are usually antifreeze compound.

11. **Cartagena protocol**
The Bio safety (Protocol to CBD) deals with;
♠ Safe handling,
♠ Storage
♠ Trans-boundary movement of the Genetically Modified Organisms (GMO).
Cartagena protocol was adopted on June 2001 in Cartagena, Spain. It entered into force on September 11th; 2003. Pakistan signed the Cartagena protocol in June 200. Pakistan has ratified it in May 2009

12. **Primary objective of category VI.** **2**
Primary objective
• To protect natural ecosystems.
• Use natural resources sustainably, when conservation and sustainable use can be mutually beneficial.

13. **Tissue culture advantage** **2**
Advantages of tissue culture conservation
• Source of disease-free material.
• It is most appropriate for rapid multiplication purposes, dissemination and active collections.

14. **Define habitat species management area.**
Definition: IUCN Management Category IV (Habitat/Species Management Area) refers to areas that are managed to protect particular species or habitats. They are defined by IUCN as *"protected areas aiming to protect particular species or habitats and management reflect this priority.*

15. **When and where Nagoya protocol held?** **2**
The protocol was adopted on 29 October 2010 in Nagoya, Japan, and entered into force on 12 October 2014. It has been ratified by 97 parties, which includes 96 UN member states and the European Union. It is the second protocol to the CBD; the first is the 2000 Cartagena Protocol on Biosafety

16. **Game reserve** **2**
"A game reserve is an area wherein controlled hunting and shooting is permitted on permit basis"

17. **Law and legislation** **2**
Laws
The system of rules which a particular country or community recognizes as regulating the actions of its members and which it may enforce by the impossible penalties.
• A person could be held guilty if he breaches Law
Legislations
The process of making or enacting laws.

18. **How genetic resource of FGR matter. (PPt 102)** **5**
The high levels of genetic variation that are present within many tree species can be beneficially developed and used by foresters and tree growers. Whereas agricultural crop breeders and farmers often substantially modify the growing environment to suit a specific crop species or variety, tree growers commonly identify species and provenances which can provide some improved levels of the goods and services required even without intensive selection and improvement, or intense management requirements, or major modification of the external environment.

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19. Future of cryopreservation. **5**

Vitrification method of cryopreservation may bring new opportunities to research protocols. It is still an experimental procedure. There are two major concern about vitrification - toxicity of high concentration of cryoprotectants used and microbial contamination of liquid nitrogen. Several IVF programs have adopted the vitrification method as the sole procedure for day-3 human embryos and for human blastocysts, with excellent survival and pregnancy rates. The challenge now is to find a protocol to successfully vitrify human oocytes for which the slow freezing method has yet to produce acceptable.

20. Historical background of CBD **5**

CBD is about the conservation and wise use of different biological resources (plants and animals).

• It was adopted in 1992 at Rio De Janeiro, Brazil and entered force on January, • 1993, which was 90 days after the 30th ratification.

• Pakistan signed it in June 1992 at United Nations Conference on Environment and Development held at Rio De Janeiro, Brazil

• Pakistan ratified it on 26th July 1994.

21. Three pillars of Ramsar

The "three pillars" of the convention

The Contracting Parties (160) commit to:

• Work towards the wise use of all their wetlands.

• Designate suitable wetlands for the list of Wetlands of International Importance.

• Cooperate internationally on transboundary wetlands, shared wetland systems and shared species.

22. Features of habitat species management area

✓ Protection of particular species

✓ Protection of habitats

✓ Active management to maintain target species

✓ Active management of culturally-defined ecosystems

23. Marine and terrestrial water act 1976

• It includes provisions for preservation, development and protection of marine environment

• This law controls marine pollution and exploration, development, conservation and management of living resources in Pakistan's Exclusive Economic Zone (EEZ)

• This law means that a ship carrying nuclear and hazardous substances will have to inform the Government of Pakistan.

24. Category_III

Definition**:** Protected areas set aside to protect a specific natural monument.

They are generally quite small protected areas and often have high visitor value

25. The services provided by PEPA

PEPA provides for;

• Protection,

• Conservation,

• Rehabilitation and

• Improvement of the environment;

• PEPA provide framework for prevention and control of pollution

• Helps in protection of sustainable development**.**

26. Strategies for effective conservation and utilization of AnGR.

Following strategies should be followed for effective conservation and utilization of AnGR:

1. Formulating the National Livestock and wild-life Breeding Policies.
2. Encouraging the Formation of Breed Associations.
3. Developing Professional Human Resources.
4. Strengthening Research and Development Institutions

Muhammad Zahid (Haveli Lakha)

5. Developing Infrastructure for marketing International co-operation and assistance is needed in capacity building to remodel available livestock farms/research stations to conserve and develop genetic resources.

27. Processing of sample in DNA bank

DNA preserved in DNA banks will be stored either within cells and extracted upon retrieval from storage or extracted from cells and purified before storage. The quality of the DNA is expressed through yield, purity, molecular weight, amplification efficiency and authenticity of sequences. The quality of DNA extracted from plant specimens is dependent on the condition of the specimen before storage, the storage environment and the duration of storage. Rapid drying of plant samples with silica gel or lyophilization helps to preserve the DNA.

28. Name the physical factors that that affect germplasm conservation

29. Application of germplasm storage

1. Seeds of many valuable species can survive long-term storage ('longterm' is defined as spanning a period of time longer than one rotation
2. Good storage facilities are now available in most of the world, and they are used extensively for tree seed storage for various regeneration purposes;
3. Seed storage is a relatively cheap method for conserving a broad range of germplasm
4. Large land areas are not tied up in conservation
5. International exchange of genetic material is facilitated by seed storag

30. Link between climate change and botanical garden

Plants can alter the temperature of the Earth's atmosphere. Through the process of photosynthesis, plants use energy from the sun to draw down carbon dioxide from the atmosphere and then use it to create the carbohydrates they need to grow. Since carbon dioxide is one of the most abundant greenhouse gases, the removal of the gas from the atmosphere may temper the warming of our planet as a whole. *transpiration in plants can increase water vapor in the atmosphere, causing more precipitation and cloud cover in an area. The additional cloud cover often reinforces the cooling by blocking sunlight. *Contribute to soil fertility and prevent soil erosion.

31. Phenotypic plasticity

Phenotypic plasticity is defined as the capacity of a particular genotype to express different phenotypes under different environmental conditions. The concept is often edztended to populations and species, with plastic trees those with flexible morphology and physiology that grow at least reasonably well under a range of different environmental stresses without genetic change. A degree of phenotypic plasticity is found in most trees, but varies substantially amongst and within species. Even in species with very little genetic diversity, such as Pinus pineaL, strong phenotypic plasticity is expressed for growth related traits, which may have helped the species colonise new environments. At least in the short term, high plasticity is likely to favour tree survival under changing environmental conditions, although trade-offs between traits can be expected. Since phenotypic plasticity has a heritable basis and may be selected for under changing environments, complex interactions between traits are possible, depending on the magnitude and structure of change.

32. Pakistan national parks

Major National parks in Pakistan are as follow;

- Hingol National Park
- Hazarganji Chiltan National Park
- Kirthar National Park
- Lal Suhanra National Park
- Margalla Hills National Park
- Ayubia National Park
- Deosai National Park
- Chitral Gol National Park

Muhammad Zahid Nawaz Lakha

- Khunjerab National Park
- Machiara National Park

33. Issues of category V

34. What are socio culture values national monuments?

Socio-cultural values

Category III areas are likely to hold socio-cultural values as they may have features such as sacred groves, springs, waterfalls, mountains, sea coves etc. of importance to one or more faith groups. areas are often of significant tourism value and can be managed with the objective of promoting sustainable tourism

35. Discuss the impact of climate change on FGR. Also discuss how this change climate effect organism which is associated with tree. 10 (Only one long FGR)

Climate change may also result in high variability in temperature and precipitation, with an increase in incidence of extreme events, such as flooding, late frosts and intensive summer droughts, amongst other events. In some areas, such as the Mediterranean and the Neo-tropics, an increase in seasonality is also expected. Under such conditions, natural selection may not result in efficient adaptation because selection pressures are multi-directional, involving traits that may be inversely correlated at the gene level. The standing genetic variation in populations may then not be large enough to create the rare new genotypic combinations that are required. Ecosystems affected by abrupt change may sustain rapid and widespread transformation as ecological tipping points are exceeded. Given the pivotal role of trees in ecosystem function, abrupt climate change impacts on them may thus have profound consequences for forests as a whole. Irreversible loss of ecosystem integrity and function may follow, with replacement by new nonendemic ecosystems.

Direct impacts of climate change

These include high tree mortality through extreme climatic events, particularly drought in combination with widespread regeneration failure, for example, examined the evidence for anthropogenic climate change leading to future large-scale “dieback” in Amazonian rain forest. Analysis suggested that dryseason water stress is likely to increase in eastern Amazonia over the 21st century, with the region tending toward a climate more appropriate to seasonal forests.

Effects of changing climate on organisms associated with trees

In particular, changes in the biology of insect pests and diseases may make ecosystems more susceptible to tree mortality. Because of improved environmental conditions for the pest and reduced tree resistance due to increased stress, pests may react to climate change with range expansions and/or increases in attack severity.

Changes in abiotic disturbance regimes

These include changes in fire regimes, flooding, landslides and/or hurricanes. Fire and climate are closely linked and are also associated with changes in land use. Coupled climate and fire-risk models suggest not only an increase in the frequency of fires but also in fire size and length of the fire-risk season, with some areas subject to risk that were not before. **Malhi et al. (2009)** considered how tipping points may be reached in Amazonian rainforest by a combination of increased dryness and an increased incidence of fire events

Invasion by organisms foreign to local ecosystems

This includes the increased risk of establishment by invasive species which accidentally arrive into ports of entry, through globalized commerce. By making new niches available, climate change will facilitate the survival of mammals, insects, diseases and/or weeds foreign to endemic ecosystems.

Muhammad Zahid (Forest Lahore)